FOR $m \geq 2$, EVERY $m \times 3$ BIQUADRATIC SUM OF SQUARES IS A SUM OF $2m$ SQUARES

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ABSTRACT. Let $\mathrm{BSR}(m,n)$ be the largest number of squares of real bilinear forms needed to write an $m \times n$ biquadratic sum of squares. We prove $\mathrm{BSR}(m,3) = 2m$ for every $m \geq 2$, answering the question of Qi, Löfberg and Chen on the exact values of $\mathrm{BSR}(m,3)$; in particular $\mathrm{BSR}(4,3) = 8$, which closes the interval $8 \leq \mathrm{BSR}(4,3) \leq 9$. The cases $m = 2, 3$ already follow from results of Blekherman, Plaumann, Sinn, Smith, Velasco and Vinzant; the new range is $m \geq 4$. The lower bound $\mathrm{BSR}(m,3) \geq 2m$ was already known, from the second order Zarankiewicz number of Löfberg and Qi; we include a short proof with an explicit form. The new part is the upper bound, which follows the boundary strategy of Blekherman, Dunbar and Sinn for ternary forms: at a hypothetical boundary point of the set of sums of $2m$ squares inside the cone, a Hankel multiplier satisfies first and second order conditions. We replace their use of Diesel's classification of Gorenstein ideals by a rank inequality for such multipliers and an induction on m, closed by a second variation built from the alternating symbol in three indices. As a consequence, $\mathrm{BSR}(m,4) \leq 3m$ for $m \geq 4$; in particular $11 \leq \mathrm{BSR}(4,4) \leq 12$.

Companion site. https://karlesmarin.github.io/blueprints/bsr-width-three/: every statement with its dependencies and its checks, and labs in the browser. This note: doi:10.5281/zenodo.24051362 (Zenodo), with its supplementary material.

1. The result

Write $x = (x_1, \ldots, x_m)$, $y = (y_1, \ldots, y_n)$; in §§2–6, $n = 3$. A *biquadratic form* is a real form of bidegree $(2,2)$ in (x,y), and $\mathrm{BSR}(m,n)$ is the least r such that every biquadratic sum of squares in $m+n$ variables is a sum of r squares of bilinear forms. It is the Pythagoras number of the coordinate ring of the Segre variety $X = \mathbb{P}^{m-1} \times \mathbb{P}^{n-1}$ in degree two [4]. By [4, Theorem 1.3], $\mathrm{BSR}(m,n) \geq mn - \mathrm{qp}(X)$; for $n = 3$ the persistence is $\mathrm{qp}(X) = m$ [7, Theorem 1], so $\mathrm{BSR}(m,3) \geq 2m$. This lower bound was already known: $\mathrm{BSR} \geq z_2$ [6, Theorem 3.4] and $z_2(m,3) = 2m$ [6]. The parallel-line covering bound of [4, Corollary 2.15] gives $\mathrm{BSR}(m,n) \leq mn - \max(m,n) + 1$ for $m, n \geq 2$ (see [7, Theorem 9]), hence $\mathrm{BSR}(2,3) = 4$ and $2m \leq \mathrm{BSR}(m,3) \leq 2m+1$ for $m \geq 3$; the theorem below removes the last unit. For $m = 4$ the lower bound 8 was first given by an explicit form of Xu, Cui and Qi [10, Theorem 4.1], and Qi, Löfberg and Chen ask for the exact values of $\mathrm{BSR}(m,3)$ [9, §9].

Theorem 1. *For every $m \geq 2$, $\mathrm{BSR}(m,3) = 2m$.*

For $m = 2$ the Segre variety $\mathbb{P}^1 \times \mathbb{P}^2$ has minimal degree and $\mathrm{BSR}(2,3) = 4$ by [3] and [4, Theorem 1.4]; for $m = 3$, $\mathbb{P}^2 \times \mathbb{P}^2$ is arithmetically Cohen–Macaulay of degree $6 = \mathrm{codim} + 2$ and $\mathrm{BSR}(3,3) = 6$ by [4, Theorem 1.5]. The statement is new for $m \geq 4$, where the new part is the upper bound. For completeness §6 gives a short proof of the lower bound by an explicit form; the rest of the note proves the upper bound.

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Strategy and attribution. The upper bound follows the boundary strategy of Blekherman, Dunbar and Sinn [2, §§2–3] (see also [3, 5]): if not every sum of squares is a sum of $2m$ squares, a generic point of the boundary of the set of sums of $2m$ squares inside the interior of the cone carries a nonzero linear functional (a Hankel multiplier) satisfying first and second order optimality conditions. In [2] the contradiction comes from Diesel's classification of height-three Gorenstein ideals; here it comes from a rank inequality for the multiplier (Lemma 5) and an induction on m whose last step is a second variation built from the alternating symbol (§5).

2. Setting

Let $A = \mathbb{R}^m$, $B = \mathbb{R}^3$ and $V = A \otimes B$, identified with the space of bilinear forms $h(x, y)$ through the monomial basis and the Euclidean inner product; $x \otimes y \in V$ is a simple tensor. Let R_2 be the space of biquadratic forms, of dimension $N = \binom{m+1}{2} \cdot 6 = 3m(m+1)$. For $k \geq 1$ put

$$\Phi_k \colon V^k \to R_2, \qquad \Phi_k(h_1, \dots, h_k) = h_1^2 + \cdots + h_k^2,$$

$\Sigma_k = \Phi_k(V^k)$ and $\Sigma = \bigcup_k \Sigma_k$, the cone of biquadratic sums of squares.

(1) $\Sigma = \Sigma_{3m}$: a form in Σ has a positive semidefinite Gram matrix of size $3m = \dim V$, hence of rank at most $3m$.
(2) Σ_k is closed: the sum of the coefficients of $x_i^2 y_j^2$ in $\Phi_k(h)$ is the sum of the squares of all coefficients of all h_i, so preimages of a bounded set are bounded.
(3) Σ is a closed convex cone with nonempty interior, since squares of bilinear forms span R_2 by polarization. A form in the interior is strictly positive at every real point of X, because evaluation at a real point is a nonzero linear functional that is nonnegative on Σ.

To $\lambda \in R_2^*$ associate the symmetric bilinear form $H_\lambda(u, v) = \lambda(uv)$ on V; $\lambda \mapsto H_\lambda$ is injective, and $H = H_\lambda$ has the *Hankel symmetry*

$$H(x \otimes b,\, x' \otimes c) = H(x \otimes c,\, x' \otimes b) \qquad (x, x' \in A,\ b, c \in B), \tag{1}$$

since both sides are λ of the same biquadratic form. In the basis of B, (1) says that the 3×3 array of $m \times m$ blocks H_{jl} of H has symmetric blocks and $H_{jl} = H_{lj}$. The differential of Φ_k is $d\Phi_k(h)(D) = 2\sum_i h_i D_i$, so

$$\lambda \circ d\Phi_k(h) = 0 \iff W_h := \langle h_1, \dots, h_k \rangle \subseteq \ker H_\lambda. \tag{2}$$

3. A regular point and a boundary point

Lemma 2. *For every $m \geq 2$ there are $h \in V^{2m}$ with integer coefficients at which $d\Phi_{2m}$ is onto. In particular Σ_{2m} has nonempty interior.*

Proof. Identify V with A^3 by the three columns (y_1, y_2, y_3). Let P be the cyclic permutation matrix of size m, $D = \operatorname{diag}(1, \dots, m)$, $Q = DP^{-1}$ and $C = [\, I \ \ P \ \ Q \,] \colon A^3 \to A$. Take for h the integer basis $(-Pe_i, e_i, 0)$, $(-Qe_i, 0, e_i)$, $1 \leq i \leq m$, of $\ker C$, so $W_h = \ker C$. By (2) we must show that a symmetric H with (1) and $\ker C \subseteq \ker H$ vanishes. Such an H is $C^\top M C$ with M symmetric, with blocks M, MP, MQ, $P^\top MP$, $P^\top MQ$, $Q^\top MQ$. Symmetry of the off-diagonal blocks gives $MP = P^\top M$, $MQ = Q^\top M$ and $P^\top MQ = Q^\top MP$. Then $P^\top MQ = P^\top Q^\top M$ and $Q^\top MP = Q^\top P^\top M$, so $(PQ - QP)^\top M = 0$, that is $M(PQ - QP) =$

0. Now $PQ - QP = PDP^{-1} - D$ is diagonal with entries $d_{\sigma(i)} - d_i = \pm 1$ or $\pm(m-1)$, hence invertible, and $M = 0$. □

Proposition 3. *Suppose $\Sigma_{2m} \neq \Sigma$. Then there are $F \in \Sigma_{2m}$ in the interior of Σ and $\lambda \in R_2^* \setminus 0$, $H = H_\lambda$, such that for every $h \in V^{2m}$ with $\Phi_{2m}(h) = F$:*

1. *$W_h \subseteq \ker H$ and $\lambda(F) = 0$;*
2. *$\sum_i H(D_i, D_i) \geq 0$ for every $D \in V^{2m}$ with $\sum_i h_i D_i = 0$;*
3. *H is indefinite;*
4. *the h_i have no common zero $([x], [y]) \in X(\mathbb{C})$.*

Proof. This is the argument of [2, Proposition 2.2 and Proposition 3.3]; we give it in our setting. Let Ω be the interior of Σ. The set $\Omega \cap \Sigma_{2m}$ contains a nonempty open set (Lemma 2 and the submersion theorem), and $\Omega \setminus \Sigma_{2m}$ is open, and nonempty because $\Sigma \setminus \Sigma_{2m}$ is a nonempty relatively open subset of $\Sigma = \overline{\Omega}$. Fix a semialgebraic stratification of R_2 compatible with Σ_{2m} and Ω and satisfying the frontier condition [1, §9]. A generic segment in the convex set Ω joining the two open sets avoids every stratum of dimension at most $N-2$ and meets the strata of dimension $N-1$ transversally; where it leaves Σ_{2m} it crosses a stratum $S \subseteq \partial\Sigma_{2m}$ of dimension $N-1$ with Σ_{2m} on one side and its complement on the other. On a small ball U around the crossing point only S and open strata meet, each component of $U \setminus S$ lies inside or outside Σ_{2m}, and there is a smooth ρ on U with $\rho = 0$ on $S \cap U$, $d\rho \neq 0$ and $\rho \geq 0$ on $\Sigma_{2m} \cap U$.

We choose $F \in S \cap U$ outside the set Z of forms in Ω that are singular at a point of $X(\mathbb{C})$. A form in Ω is positive on $X(\mathbb{R})$, and a singular point $p = ([x], [y])$ has no real factor: if $[x]$ is real, take a real representative $x \neq 0$ and write $F(x, y) = y^{\mathsf{T}} M_F(x)\, y$; since F is positive on $X(\mathbb{R})$, $M_F(x)$ is positive definite, and singularity at p would require $M_F(x)y = 0$, impossible also over $\mathbb{C}$ (symmetrically if $[y]$ is real). So p and $\bar{p}$ differ in both factors. After complex changes of coordinates $p, \bar{p}$ become $(e_1, f_1), (e_2, f_2)$; singularity at (e_i, f_i) forces to vanish, in the chart $x_i = y_i = 1$, the coefficients of $x_i^2 y_i^2$, $x_i x_j y_i^2$ $(j \neq i)$ and $x_i^2 y_i y_l$ $(l \neq i)$, which are $m+2$ conditions, and the two lists are disjoint. For real forms this is $2(m+2)$ real conditions, while p moves in $2(m+1)$ real dimensions; hence $\dim Z \leq N - 2 < \dim S$.

Put $\lambda = d\rho_F$. For a preimage h, $\rho \circ \Phi_{2m} \geq 0$ near h and vanishes at h, so it has a local minimum there: the first order condition is $\lambda \circ d\Phi_{2m}(h) = 0$, which is (1) by (2), and $\lambda(F) = \sum_i H(h_i, h_i) = 0$. For D in the kernel of $d\Phi_{2m}(h)$ the second derivative of $\rho \circ \Phi_{2m}$ along D is $d\rho_F(d^2\Phi_{2m}[D, D]) = 2\sum_i H(D_i, D_i)$, which is nonnegative at a minimum: this is (2). If H or $-H$ were positive semidefinite, λ or $-\lambda$ would be nonnegative on Σ and vanish at the interior point F, so $\lambda = 0$; this proves (3). A common complex zero of the h_i is a singular point of F, excluded by $F \notin Z$: this is (4). □

Corollary 4. *In the situation of Proposition 3, for every representation $F = \sum_{i \leq 2m} h_i^2$ the h_i are linearly independent, F is not a sum of fewer than $2m$ squares, $r := \operatorname{rank} H \leq m$, and $\operatorname{im} H_{\mathbb{C}}$ contains no nonzero simple tensor $x \otimes y$.*

Proof. This is [2, Lemma 3.4 and Proposition 3.3(4)] in our setting. If the h_i were dependent, an orthogonal change of the $2m$ summands (which does not change F) would make $h_{2m} = 0$; then $D = (0, \ldots, 0, v)$ satisfies $\sum h_i D_i = 0$ for every v, and Proposition 3(2) gives $H(v, v) \geq 0$ for all v, against (3). A representation with fewer squares, completed with zeros, would be dependent. So $\dim W_h = 2m$ and $\operatorname{im} H \subseteq W_h^{\perp}$ has dimension at most $3m - 2m = m$. Finally $\operatorname{im} H_{\mathbb{C}} \subseteq (W_h)^{\perp}_{\mathbb{C}}$, and $x \otimes y \in (W_h)^{\perp}_{\mathbb{C}}$ means $h_i(x, y) = 0$ for all i, excluded by (4). □

4. The support of a multiplier

For a symmetric H with (1) let $A'' = \{x \in A :\ H(x \otimes b, v) = 0 \text{ for all } b \in B,\ v \in V\}$ and $a = \dim A - \dim A''$: the support of H in the first factor has dimension a, and H induces on $A' \otimes B$, for any complement A' of A'', a form of the same rank with trivial such radical.

Lemma 5. *Let $H \neq 0$ be symmetric with (1) and support of dimension a in the first factor. If $\operatorname{im} H_{\mathbb{C}}$ contains no nonzero simple tensor, then $\operatorname{rank} H \geq a + 1$.*

Proof. Restrict to $A' \otimes B$, complexify and write $H(u, v) = J(Tu, Tv)$ with $T\colon A'_{\mathbb{C}} \otimes B_{\mathbb{C}} \to E$ onto, $\dim E = r = \operatorname{rank} H$, and J a nondegenerate symmetric bilinear form on E. Put $T_b x = T(x \otimes b)$. Trivial radical means $\bigcap_b \ker T_b = 0$, and (1) reads

$$J(T_b x, T_c v) = J(T_c x, T_b v). \tag{3}$$

A line gives a simple tensor. If $x \neq 0$ and $T_b x = \beta(b)e$ for a linear form $\beta \neq 0$ and $e \neq 0$, choose c with $\beta(c) = 1$; by (3), $J(e, T_b v) = \beta(b)\, J(e, T_c v)$ for all $v \in A'$, $b \in B$. The left side, as a linear form in $v \otimes b$, lies in $\operatorname{im} H_{\mathbb{C}}$ (T is onto, J nondegenerate) and is nonzero; the right side is the simple tensor $\beta \otimes J(e, T_c\,\cdot)$. So it suffices to find such an x when $r \leq a$.

Assume $r \leq a$. Choose b with $s = \operatorname{rank} T_b$ maximal, $U = \operatorname{im} T_b$, $N_0 = \ker T_b$. For every c,

$$T_c(N_0) \subseteq U \cap U^{\perp_J}. \tag{4}$$

Orthogonality follows from (3) with $x \in N_0$. For $T_c(N_0) \subseteq U$, write T_b as an identity block of size s and zeros; the coefficient of t in an $(s+1)$-minor of $T_b + tT_c$ bordering that block is an entry of $T_c\colon N_0 \to E/U$, and maximality of s forces it to vanish.

If $s = r < a$, then $U = E$, $U \cap U^{\perp_J} = 0$, and (4) gives a common kernel $N_0 \neq 0$, a contradiction. If $s = r = a$, T_b is invertible; complete b to a basis b, c, d. The operators $T_b^{-1}T_c$, $T_b^{-1}T_d$ are self-adjoint for $J(T_b\,\cdot, T_b\,\cdot)$ by (3), and (3) for the pair c, d makes their product self-adjoint as well, so they commute; a common eigenvector $x \neq 0$ gives $T_c x, T_d x \in \mathbb{C}\, T_b x$, a line. If $s < r$, put $R = U \cap U^{\perp_J}$; then $\dim N_0 = a - s \geq r - s \geq \dim R$. If $R = 0$ there is again a common kernel. Otherwise $T_c|_{N_0}, T_d|_{N_0}\colon N_0 \to R$ form a pencil: if $\dim N_0 > \dim R$ every member has a kernel, and if the dimensions agree the binary form $\det(\sigma T_c + \tau T_d)|_{N_0}$ is zero or of positive degree, so it vanishes at some $[\sigma : \tau] \in \mathbb{P}^1(\mathbb{C})$. Either way there is $x \in N_0 \setminus 0$ with $T_b x = 0$ and $T_c x, T_d x$ dependent, not both zero; they span a line. In all cases the first paragraph gives a contradiction. □

5. Induction and second variation

Proof of the upper bound in Theorem 1. We prove $\Sigma_{2m} = \Sigma$ by strong induction on $m \geq 2$. Suppose not, and take F, H as in Proposition 3; fix a representation $F = \sum_{i \leq 2m} h_i^2$ and $W = W_h$. By Corollary 4 and Lemma 5, $a + 1 \leq r \leq m$; as $H \neq 0$, $a \geq 1$, so $1 \leq a \leq m - 1$. Let $A = A' \oplus A''$ orthogonally, $e = m - a$.

Case $e \geq 2$. Write $w = w' + w''$ with $w' \in A' \otimes B$, $w'' \in A'' \otimes B$. Since $A'' \otimes B$ lies in the radical of H and $W \subseteq \ker H$, the projection W' of W to $A' \otimes B$ lies in the kernel of $H|_{A' \otimes B}$, which has dimension $3a - r$. An orthogonal change of the $2m$ summands moves all the components in $A' \otimes B$ into at most $s = \dim W' \leq 3a - r$ of them; the other $2m - s$ summands are bilinear forms in the e variables of A'', and their sum of squares is a sum of squares in $e \times 3$ variables. As $2 \leq e < m$, by induction it is a sum of $2e$ squares. Hence F is a sum of at most $3a - r + 2(m - a) = 2m + a - r \leq 2m - 1$ squares, against Corollary 4.

Case $e = 1$. Then $a = m-1$ and $r = m = a+1$, so $W = \ker H$ and $A'' \otimes B \subseteq W$; let x_0 span A''. The coefficient map $\mathbb{R}^{2m} \to W$ of the representation is an isomorphism (Corollary 4); choosing an orthonormal basis of the preimage of $x_0 \otimes B$ and completing it, an orthogonal change of the summands gives $h_i = x_0 \otimes y_i$ $(i = 1, 2, 3)$ for a basis y_1, y_2, y_3 of B. Write H_{jl} for the blocks of $H|_{A' \otimes B}$ in this basis (symmetric, with $H_{jl} = H_{lj}$ by (1)) and $S = H_{11} + H_{22} + H_{33}$. For $v_1, v_2, v_3 \in A'$ let

$$D_i = \sum_{j,p} \varepsilon_{ijp}\, v_p \otimes y_j \quad (i = 1, 2, 3), \qquad D_i = 0 \quad (i > 3),$$

with ε the alternating symbol. Then $\sum_i h_i D_i = \sum_{i,j,p} \varepsilon_{ijp}(x_0 \otimes y_i)(v_p \otimes y_j) = 0$ by antisymmetry in i, j, and $\sum_i \varepsilon_{ijp}\varepsilon_{ilq} = \delta_{jl}\delta_{pq} - \delta_{jq}\delta_{pl}$ gives

$$\sum_i H(D_i, D_i) = \sum_p v_p^\top S v_p - \sum_{p,q} v_p^\top H_{pq} v_q.$$

By Proposition 3(2) this is nonnegative for all v, that is $L := I_3 \otimes S - H|_{A' \otimes B} \succeq 0$ in the block order. The three diagonal blocks of L sum to $3S - S = 2S$, so $S \succeq 0$. If $Sx = 0$, the three nonnegative numbers $L(x \otimes y_j, x \otimes y_j)$ sum to $2x^\top S x = 0$, so $L(x \otimes y_j, \cdot) = 0$ and hence $H(x \otimes y_j, \cdot) = 0$ for all j; by the choice of A', $x = 0$. So $S \succ 0$. The congruence by $\operatorname{diag}(S^{-1/2}, S^{-1/2}, S^{-1/2})$ turns $H|_{A' \otimes B}$ into $\widetilde{H}$ with $\widetilde{H} \preceq I_{3a}$, $\operatorname{tr} \widetilde{H} = \operatorname{tr}(S^{-1/2} S S^{-1/2}) = a$, $\operatorname{rank} \widetilde{H} = a + 1$ and the same positive and negative inertia indices as H, which by Proposition 3(3) has a negative eigenvalue. Then $\widetilde{H}$ has at most a positive eigenvalues, each at most 1, and a negative one, so $\operatorname{tr} \widetilde{H} < a$: a contradiction.

For $m = 2$ only $e = 1$ occurs, so the induction starts without hypothesis. □

6. The lower bound

Lemma 6. *The form* $F_m = \sum_{i=1}^m x_i^2\big((y_2 - iy_1)^2 + y_3^2\big)$ *is a sum of* $2m$ *squares of bilinear forms and of no fewer.*

Proof. The displayed expression has $2m$ squares. Write any representation as

$$F_m = \Big\| \sum_i x_i M_i y \Big\|^2, \qquad M_i \in \mathbb{R}^{s \times 3}.$$

The coefficient of x_i^2 gives $M_i^\top M_i$ of rank two with kernel spanned by $b_i = (1, i, 0)$. For $i \neq j$ the coefficient of $x_i x_j$ vanishes, so $M_i^\top M_j$ is antisymmetric; it kills b_j on the right and b_i on the left, hence, being antisymmetric, b_i and b_j on the right. An antisymmetric 3×3 matrix with two independent vectors in its kernel is zero. So the images of the M_i are m planes, pairwise orthogonal in $\mathbb{R}^s$, and $s \geq 2m$. □

This gives $\mathrm{BSR}(m, 3) \geq 2m$, which was already known from $z_2(m, 3) = 2m$ [6] and also follows from $\mathrm{qp}(\mathbb{P}^{m-1} \times \mathbb{P}^2) = m$ and [4, Theorem 1.3]; with §5 it completes the proof of Theorem 1. □

7. Larger widths

Corollary 7. *For* $m \geq 4$, $\mathrm{BSR}(m, 4) \leq 3m$. *In particular* $11 \leq \mathrm{BSR}(4, 4) \leq 12$.

Proof. We prove more generally $\mathrm{BSR}(m,n) \le mn - \max(m,n)$ for $m,n \ge 3$; for $n = 4 \le m$ this is $3m$. This is the argument of [7, Theorem 9] with Theorem 1 in place of width two. Let $m \ge n$ and $f = \sum_{k \le r} b_k^2$; let $v_{ij} \in \mathbb{R}^r$ be the coefficient vector of $x_i y_j$ in $(b_1, \dots, b_r)$. The vectors v_{ij} with $j \ge 4$ span a space W of dimension at most $m(n-3)$. After an orthogonal change of the r coordinates, which does not change f, W lies in the first $\dim W$ of them, and the remaining coordinates give bilinear forms in x and y_1, y_2, y_3 only. Their sum of squares is an $m \times 3$ biquadratic sum of squares, hence a sum of $2m$ squares by Theorem 1. So f is a sum of $m(n-3) + 2m = mn - m$ squares. For $m < n$ exchange the roles of x and y. The lower bound $\mathrm{BSR}(4,4) \ge 11$ is [8, Theorem 3.1]. □

In width four the previous upper bound was the covering bound $3m+1$, which for $m = 4, 5, 6$ is also the bound of [4, Theorem 2.1] (13 for $m = 4$). For $\min(m,n) = 3$ the general bound is the exact value $2\max(m,n)$ of Theorem 1, also for $m = 3$ and n large, and for $m = n = 3$ it is the known value 6. For $\min(m,n) \ge 5$ it is one less than the covering bound but never better than [4, Theorem 2.1], which gives $\binom{\mathrm{BSR}+1}{2} < \binom{m+1}{2}\binom{n+1}{2}$: with $s = \min(m,n)$, $M = \max(m,n)$ and $C = M(s-1)$,

$$4\Big(\tbinom{C+2}{2} - \tbinom{M+1}{2}\tbinom{s+1}{2}\Big) = (M-1)\big(M(s^2 - 5s + 2) - 4\big),$$

which is positive for $s \ge 5$ (so that bound is at most C) and negative for $s = 4$ (so it is at least $C+1$). Both give 20 for 5×5, and for 6×6 that bound gives 29 against 30. Figure 1 shows the three bounds in widths four and five.

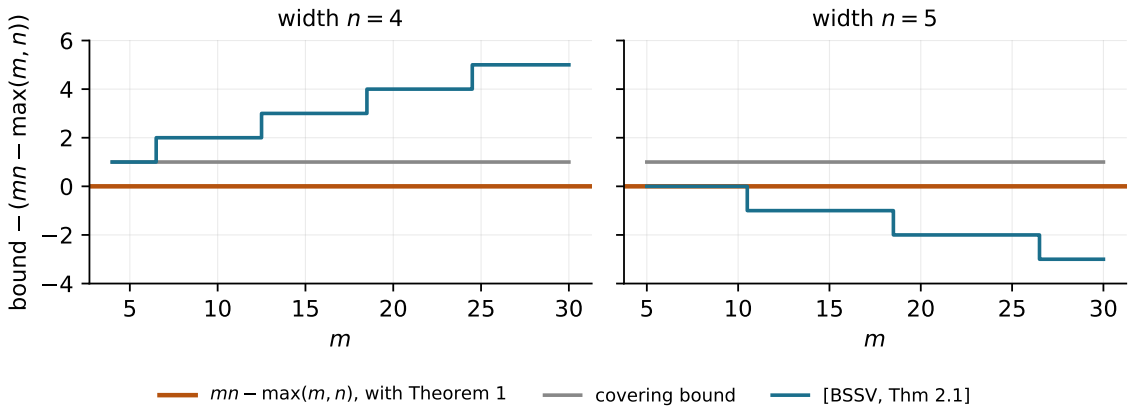


FIGURE 1. Upper bounds for $\mathrm{BSR}(m,n)$ minus $mn - \max(m,n)$, the bound obtained with Theorem 1: the covering bound of [4, Corollary 2.15] and the bound of [4, Theorem 2.1]. In width four both lie above for every $m \ge 4$ (Corollary 7); in width five, the BSSV bound equals $mn - \max(m,n)$ for $5 \le m \le 10$ and is lower for every $m \ge 11$.

8. Checks and formal status

The regular-point ranks in Lemma 2 were checked modulo 1 000 003 for $2 \le m \le 7$. For $m = 4$, the control $Q = P$ gave rank 56 out of 60. The identity in §5 was tested numerically on 100 random Hankel forms in non-orthonormal bases. For $m = 5$, exact arithmetic verified

the existence of a Hankel multiplier $H = C^\top JC$ of rank 5 and support of dimension 4 whose complex image contains no simple tensor (the Hankel forms $C^\top EC$ form a line, since a 14×14 minor of the linear conditions equals -48; a simple tensor $x \otimes y = C^\top z$ in the image would give a second one, $C^\top zz^\top C$). The example attains equality in Lemma 5 under its hypotheses. Separately, a numerical search found approximate decompositions with $2m$ squares for every sampled $m \times 3$ form with $m = 4, 5, 6$, and none with $2m - 1$ for F_m; for F_m the latter is proved exactly by Lemma 6. The scripts accompany this note, and on the companion site the rank at the regular point and the $2m$ squares of F_m are computed in the reader's browser. Part of the argument is checked in Isabelle/HOL, without `sorry`, in the public repository of formalizations: the lower bound $\mathrm{BSR}(m, 3) \geq 2m$ of Lemma 6 (every representation of F_m has at least $2m$ squares), the algebraic core of Lemma 2 (the three block symmetries force $M = 0$ for P cyclic and $Q = DP^{-1}$), and the case $e = 1$ of §5 (the second variation identity and the trace bound, stated with rank at most $a + 1$). Lemma 5, the case $e \geq 2$, the passage from the Hankel multiplier to the block symmetries, Proposition 3 and the assembly of Theorem 1 are not formalized.

The companion site

This note has a companion website, its *blueprint*, at `https://karlesmarin.github.io/blueprints/bsr-width-three/`. There one finds:

- *The statements.* Theorem 1, Lemmas 2, 5 and 6, Proposition 3 and Corollaries 4 and 7, copied verbatim with their numbers, each with its proof and a commentary in plain words.
- *The dependencies.* What each statement uses and where it is used, as a list and as a graph.
- *The map.* The result, the tools it rests on and the question it answers.
- *Labs in the browser.* The Gram spectrahedron of a random form with a Gram matrix of rank $2m$ on its boundary; the matrices of Lemma 2 and the diagonal of $PQ - QP$; the form F_m with its $2m$ squares; the three upper bounds of Figure 1 for any m and n.
- *Formal proofs.* For each statement, which part is checked in Isabelle/HOL and which is not.
- *The note itself.* Its PDF, opened at the right page from every statement.

Questions

Does the method give $\mathrm{BSR}(m, 4)$? For $m \geq 4$, our earlier note and Corollary 7 give $\lfloor 8m/3 \rfloor \leq \mathrm{BSR}(m, 4) \leq 3m$. In width four, if the kernel of the multiplier H contains k linearly independent summands, its rank is at most $4m - k$, and the second variation would have to come from a structure replacing the alternating symbol on $\mathbb{R}^3$.

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